

Maze Hops

CITS3001 Advanced Algorithms — Time limit: 3s

You are lost in the CSSE building, which is best modelled as a rectangular maze. Each cell is either open (.) or a wall (#). From an open cell you can step to any of the four adjacent open cells (up, down, left, right). Find the minimum number of steps to get from the start **S** to the exit **E**.

Input

The first line contains two integers R and C ($1 \leq R, C \leq 500$), the number of rows and columns. Each of the next R lines contains a string of C characters from $\{., \#, S, E\}$. There is exactly one **S** and exactly one **E**; both are open cells.

Output

Output the minimum number of steps from **S** to **E**, or -1 if the exit cannot be reached.

Sample Input 1

```
3 4
S..#
.#..
..E.
```

Sample Output 1

```
4
```

Sample Input 2

```
2 2
S#
#E
```

Sample Output 2

```
-1
```